Title:

AI/ML Task-2: Feature Engineering, Model Optimization & Performance Comparison

1️⃣ Introduction

This project focuses on applying machine learning regression models to predict housing prices using the California Housing dataset. The objective is to perform feature engineering, apply multiple regression models, optimize performance, and compare results using evaluation metrics.

2️⃣ Dataset Description

The project uses the California Housing dataset available in scikit-learn. The dataset contains features such as:

- Median Income
- House Age
- Average Rooms
- Average Bedrooms
- Population
- Average Occupancy
- Latitude
- Longitude

The target variable represents median house value.

3️⃣ Methodology

The following steps were performed:

1. Imported necessary Python libraries.
2. Loaded the California Housing dataset.
3. Performed feature inspection using pandas.
4. Split data into training and testing sets (80%-20%).
5. Applied feature scaling using StandardScaler.
6. Implemented three regression models:

- Linear Regression
- Ridge Regression (for regularization)
- Decision Tree Regressor

7. Evaluated models using:

- RMSE (Root Mean Squared Error)
- R^2 Score

8. Compared model performances in tabular format.

4️⃣ Results

The model performance comparison is shown below:

(You can paste screenshot of your comparison table here)

Analysis:

- Ridge Regression slightly improved over Linear Regression due to regularization.
- Decision Tree showed competitive performance but may overfit.

- Feature scaling improved linear model accuracy.

5️⃣ Conclusion

In this project, multiple regression models were implemented and compared. Ridge Regression demonstrated improved generalization compared to simple Linear Regression. The evaluation metrics RMSE and R^2 provided clear insight into model accuracy. Overall, proper feature scaling and model optimization significantly improved performance